Koji Andriamahery

SYSTEMS ENGINEER, HEALTH TECHNOLOGIES

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Overview

With an emphasis on R&D, my interests promote cross-disciplinary engineering in the medical device industry. My background mainly lies in **active medical device design,** from **signal processing** and **firmware development** to **mechatronics** and **test automation.** Familiar with **ISO 13485, ISO 14971, IEC 62304, IEC 60601** and **IEC 62366.**

Work Experiences

DeepPulseNeuro — Montpellier, France
R&D Engineer, Neurophysiology — 08/2025 - current

- Embedded systems design, including PCB design, firmware programming (**C/C++**, **Matlab**) and test automation (**Python**).
- Implemented neurovascular phantom for mechanical and electrical testing (patent pending) of medical device prototypes.

Erba Diagnostics France (R&D centre of Erba Mannheim) — Montpellier, France
Electrical Engineer, Hematology — 04/2024 - 08/2025

- Contributed to IVD hematology analyzers system design, with a focus on blood cell differentiation (Complete Blood Count).
- Worked on analytic performance optimisation, embedded software development (**C, C#**) and systems fine tuning - including hands-on assemblies of proof-of-concepts in close collaboration with biologists, mechanical and software engineers.

Stella Surgical — Montpellier, France
R&D Engineer, Embedded systems for graft monitoring — 01/2022 - 04/2024

- Led process modeling (**MBSE**) and embedded software development, including firmware implementation (**C, C++**), test automation (**Python**) and cloud backend integration (**TypeScript**) for organ tracking IoT sensors.
- Initiated and designed a physical perfusion system simulator from scratch for surgical training and demonstrations.

LIRMM - Smart Integrated Electronic Systems Group — Montpellier, France
Research Engineer, Embedded physiological sensing — 09/2021 - 01/2022

- Initiated a personal research experimentation : Embedded hand gesture recognition for sign language interpretation.
- Led hardware architecture, PCB design (wearable form-factor), signal processing (**C/C++, Matlab**) and edge ML.

HumanLab Saint-Pierre (Children's hospital) — Palavas-Les-Flots, France
Assistive Technology Designer, Rehabilitation Technologies — 01/2021 - 07/2021

- Developed a wearable assistive robotic orthosis (**C/C++**) for children, in collaboration with patients and health care professionals (physicians, prosthetists /orthotists, occupational therapists) from Institut Saint-Pierre (paediatric hospital).

TU Delft - Interactive Intelligence Group — Delft, The Netherlands
Research Assistant, Responsible AI — 05/2020 - 08/2020

- Applied data-analysis techniques (**Python, SQL**) on MIT's "Moral Machine" dilemma dataset to build ethical decision-making frameworks that incorporate moral uncertainty on autonomous vehicle systems, with peer-reviewed publication.

Education

University of Montpellier - Faculty of Medicine and Sciences. — 2022
MS in Digital Health - Minor : Health Devices Engineering

- Graduated with high honors (Ranked 1st).
- Thesis : IoT-based system design for clinical decision support and graft monitoring during organ transplantation processes.

IMT Mines Alès - Engineering School. — 2022
Diplôme d'ingénieur généraliste (MEng) - Major : Mechatronics

- Lead organizer and First TEDx Licensee of both the Engineering School and the city.
- Research track on mobile robotics (sensor fusion, embedded systems and control).

Miscellaneous

Languages : French (native), English (fluent, C2 level - CEFR guideline), Spanish (notions).

Music : Graduated in jazz music from Conservatory of Music and Dance in 2018 (Amiens, France) - instruments : saxophone. bass guitar, drums, EWI, and some weird sounding homemade prototyped things.